HDOC Day-74

Enumerations in C: Months and Operation Status

Problem Statement

1. Create an enumeration for months and print how many days each month has. Example: FEB -> 28 or 29 days.

2. Define an enumeration with SUCCESS, FAILURE, and TIMEOUT, and print appropriate messages. Example: FAILURE -> Operation Failed.

Algorithm

1. Define enum Month with JAN through DEC and enum Status with SUCCESS, FAILURE, TIMEOUT.

2. Repeatedly display a menu: months, status, exit.

3. For months, read a three-letter abbreviation and convert it to uppercase; reject unknown months.

4. If FEB is selected, read a year and apply the leap-year rule (divisible by 400, or divisible by 4 but not 100). Otherwise select 30 or 31 days as appropriate.

5. For status, read a name, match it to the enum value, and display the message using switch.

6. Handle invalid menu choices; exit when choice 3 is selected.

Test Case Table

Test Case	Menu Choice	Additional Input	Expected Output
1	1	JAN	JAN has 31 days.
2	1	FEB; year 2024	FEB has 29 days.
3	1	FEB; year 2023	FEB has 28 days.
4	1	APR	APR has 30 days.
5	1	FEB; year 1900	FEB has 28 days.
6	1	FEB; year 2000	FEB has 29 days.
7	2	SUCCESS	Operation Successful
8	2	FAILURE	Operation Failed
9	2	TIMEOUT	Operation Timed Out
10	9, then 3	Invalid choice, then exit	Invalid choice. Enter 1 to 3.; Exiting program.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

enum Month { JAN=1, FEB, MAR, APR, MAY, JUN, JUL, AUG, SEP, OCT, NOV, DEC };
enum Status { SUCCESS, FAILURE, TIMEOUT };

void monthDays(void) {
    const char *names[] = {"JAN","FEB","MAR","APR","MAY","JUN","JUL","AUG","SEP","OCT","NOV","DEC"};
```

```
    char input[40];
    int i, year, days;
    printf("Enter month abbreviation (JAN to DEC): ");
    if (scanf("%39s", input) != 1) return;
    for (i=0; input[i]; i++) input[i]=(char)toupper((unsigned char)input[i]);
    for (i=0; i<12; i++) if (strcmp(input,names[i])==0) break;
    if (i==12) { printf("Invalid month.\n"); return; }
    enum Month month=(enum Month)(i+1);
    if (month==FEB) {
        printf("Enter year: ");
        if (scanf("%d", &year)!=1 || year<1) { printf("Invalid year.\n"); return; }
        days=((year%400==0)||(year%4==0 && year%100!=0))?29:28;
    } else if (month==APR || month==JUN || month==SEP || month==NOV) days=30;
    else days=31;
    printf("%s has %d days.\n", names[i], days);
}

void statusMessage(void) {
    char input[40];
    int i;
    printf("Enter status (SUCCESS/FAILURE/TIMEOUT): ");
    if (scanf("%39s", input)!=1) return;
    for (i=0; input[i]; i++) input[i]=(char)toupper((unsigned char)input[i]);
    enum Status status;
    if (strcmp(input,"SUCCESS")==0) status=SUCCESS;
    else if (strcmp(input,"FAILURE")==0) status=FAILURE;
    else if (strcmp(input,"TIMEOUT")==0) status=TIMEOUT;
    else { printf("Invalid status.\n"); return; }
    switch (status) {
        case SUCCESS: printf("Operation Successful\n"); break;
        case FAILURE: printf("Operation Failed\n"); break;
        case TIMEOUT: printf("Operation Timed Out\n"); break;
    }
}

int main(void) {
    int choice;
    while (1) {
        printf("\n--- HDOC DAY-74 MENU ---\n1. Days in a month\n2. Operation status\n3. Exit\nEnter choice: ");
        if (scanf("%d", &choice)!=1) { printf("Invalid numeric input.\n"); return 1; }
        switch(choice) {
            case 1: monthDays(); break;
            case 2: statusMessage(); break;
            case 3: printf("Exiting program.\n"); return 0;
            default: printf("Invalid choice. Enter 1 to 3.\n");
        }
    }
}
```

Compilation and Execution

gcc hdoc74.c -o hdoc74

./hdoc74

Compilation and Execution



Test Case 1

Menu choice: 1 | Input: JAN

Expected: JAN has 31 days.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc74

── HDOC DAY-74 MENU ──
1. Days in a month
2. Operation status
3. Exit
Enter choice: 1
Enter month abbreviation (JAN to DEC): JAN
JAN has 31 days.
```

Test Case 2

Menu choice: 1 | Input: FEB; year 2024

Expected: FEB has 29 days.

```
── HDOC DAY-74 MENU ──
1. Days in a month
2. Operation status
3. Exit
Enter choice: 1
Enter month abbreviation (JAN to DEC): FEB
Enter year: 2024
FEB has 29 days.
```

Test Case 3

Menu choice: 1 | Input: FEB; year 2023

Expected: FEB has 28 days.

```
── HDOC DAY-74 MENU ──
1. Days in a month
2. Operation status
3. Exit
Enter choice: 1
Enter month abbreviation (JAN to DEC): FEB
Enter year: 2023
FEB has 28 days.
```

Test Case 4

Menu choice: 1 | Input: APR

Expected: APR has 30 days.

```
── HDOC DAY-74 MENU ──
1. Days in a month
2. Operation status
3. Exit
Enter choice: 1
Enter month abbreviation (JAN to DEC): APR
APR has 30 days.
```

Test Case 5

Menu choice: 1 | Input: FEB; year 1900

Expected: FEB has 28 days.

```
── HDOC DAY-74 MENU ──
1. Days in a month
2. Operation status
3. Exit
Enter choice: 1
Enter month abbreviation (JAN to DEC): FEB
Enter year: 1900
FEB has 28 days.
```

Test Case 6

Menu choice: 1 | Input: FEB; year 2000

Expected: FEB has 29 days.

```
── HDOC DAY-74 MENU ──
1. Days in a month
2. Operation status
3. Exit
Enter choice: 1
Enter month abbreviation (JAN to DEC): FEB
Enter year: 2000
FEB has 29 days.
```

Test Case 7

Menu choice: 2 | Input: SUCCESS

Expected: Operation Successful

```
── HDOC DAY-74 MENU ──
1. Days in a month
2. Operation status
3. Exit
Enter choice: 2
Enter status (SUCCESS/FAILURE/TIMEOUT): SUCCESS
Operation Successful
```

Test Case 8

Menu choice: 2 | Input: FAILURE

Expected: Operation Failed

```
── HDOC DAY-74 MENU ──
1. Days in a month
2. Operation status
3. Exit
Enter choice: 2
Enter status (SUCCESS/FAILURE/TIMEOUT): FAILURE
Operation Failed
```

Test Case 9

Menu choice: 2 | Input: TIMEOUT

Expected: Operation Timed Out

```
── HDOC DAY-74 MENU ──
1. Days in a month
2. Operation status
3. Exit
Enter choice: 2
Enter status (SUCCESS/FAILURE/TIMEOUT): TIMEOUT
Operation Timed Out
```

Test Case 10

Menu choice: 9, then 3 | Input: Invalid choice, then exit

Expected: Invalid choice. Enter 1 to 3.; Exiting program.

```
── HDOC DAY-74 MENU ──
1. Days in a month
2. Operation status
3. Exit
Enter choice: 9
Invalid choice. Enter 1 to 3.

── HDOC DAY-74 MENU ──
1. Days in a month
2. Operation status
3. Exit
Enter choice: 3
Exiting program.
```